

Federated Retraction Prediction: Scaling Scientific-Integrity Screening Across Publisher Silos

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Abstract

Retractions of scientific publications are growing rapidly, straining the integrity of the scholarly record, yet automated screening remains limited by two structural constraints: prior work relies on small hand-curated corpora, and the strongest predictive signals (full text, citation contexts, and editorial metadata) are fragmented across publishers that cannot legally pool their data. We cast large-scale retraction prediction as a *cross-silo federated learning* problem. From the Retraction Watch database enriched with OpenAlex metadata we construct a corpus of 138,537 articles (37,648 retracted; publisher- and year-matched controls), $180\times$ larger than the largest corpus in prior retraction-prediction studies. We simulate realistic publisher and discipline silos, naturally non-IID in size, era, and topical composition, and benchmark local-only and centralized training against FedAvg and FedProx, adding a differentially private variant. Federated training attains 92% of the AUPRC of an identical centrally trained model (0.558 vs. 0.609) while improving 77% over the silo-local status quo (0.315), and remains stable as the

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consortium grows from 5 to 20 silos. Cross-corpus experiments against the hand-curated benchmark of prior work show that models trained on hundreds of curated articles collapse to chance on the open corpus, that abstract-level features alone reduce prior reported accuracy from 0.88 to chance (quantifying how much signal is locked in paywalled full text), and that linguistic features transfer across domains while metadata features do not. Together these results position federated learning as the deployment path for retraction screening at the scale of the scientific record.

Keywords: Retraction prediction, Federated learning, Scientific integrity, Digital libraries, Scholarly communication, Differential privacy

1. Introduction

The retraction of scientific publications has grown from isolated incidents to a systemic phenomenon: the Retraction Watch database now records more than 71,000 retraction events, with over 15,000 in 2024 alone [1, 2]. Because a retraction takes up to three years on average to be issued, flawed results continue to be cited, reused, and built upon long after problems are first suspected [3]. For digital libraries, indexing services, and publishers, the ability to flag at-risk articles early is therefore a core problem of maintaining the integrity of the scholarly information ecosystem. A line of recent work by Usman and Balke [3, 4, 5, 6] demonstrated that retraction is predictable from how articles are written and cited: readability and certainty of full-text sections separate retracted from non-retracted articles with 0.88 accuracy [6], and citation sentences anticipate retraction cascades [4].

Yet this evidence was obtained on corpora of a few hundred hand-curated

articles, and its authors note that re-evaluating the complete scientific literature with such methods is infeasible. We argue the obstacle is structural rather than methodological. The strongest features require full text and editorial context, which are held by *different publishers* and cannot be pooled for legal and commercial reasons. Any deployable system must therefore learn *across* data silos without moving the data.

Problem statement.. Let $\mathcal{P} = \{P_1, \dots, P_K\}$ be K data owners (publishers or disciplinary databases). Each holds a private dataset $D_k = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n_k}$ of articles with features $\mathbf{x}_i \in \mathbb{R}^d$ and retraction labels $y_i \in \{0, 1\}$, where the D_k differ in size, feature distribution, and label prevalence (statistical heterogeneity). The goal is to learn a global predictor f_θ minimizing

$$\min_{\theta} F(\theta) = \sum_{k=1}^K \frac{n_k}{n} F_k(\theta), \quad F_k(\theta) = \frac{1}{n_k} \sum_{(\mathbf{x}, y) \in D_k} \ell(f_\theta(\mathbf{x}), y), \quad (1)$$

subject to the constraint that raw records never leave their silo:

$$D_k \text{ is accessed only by } P_k, \quad \text{only model parameters } \theta \text{ are exchanged.} \quad (2)$$

Contributions.. (1) The first formulation of retraction prediction as cross-silo federated learning, with an open, reproducible benchmark of 138,537 articles across naturally non-IID publisher and discipline silos, $180\times$ larger than prior retraction-prediction corpora. (2) A systematic comparison of local-only, centralized, and federated training (FedAvg, FedProx), showing federation recovers 92% of centralized AUPRC and improves 77% over the silo-local status quo, robustly across two silo definitions, consortium sizes 5–20, and a temporal split. (3) A differential-privacy analysis quantifying the utility cost

of protecting individual records in the federated setting. (4) Cross-corpus experiments against the benchmark of [6] that measure, for the first time, how much retraction signal is locked in paywalled full text, and reveal that linguistic features transfer across domains while metadata features do not.

2. Related Work

Retraction analysis and prediction. Usman and Balke introduced citation-intention analysis as a quality-control mechanism [3], traced retraction cascades to identify not-yet-retracted but suspicious articles [4], defined *concerning citations* as early self-correction signals [5], and showed that section-level readability and certainty separate methodology-error retractions from matched controls with Random Forests at 0.88 accuracy, outperforming BioBERT [6]. Within this journal, bibliometric studies similarly identify interpretable metadata correlates of retraction [7]. **Limitation.** All rely on centralized corpora of at most a few thousand hand-curated instances; the features that perform best require full text that no single party can obtain at scale.

Federated learning. FedAvg [8] established communication-efficient averaging of locally trained models; FedProx [9] adds a proximal term for heterogeneous clients; a broad survey is given in [10]. Differentially private aggregation clips and noises client updates [11]. Cross-silo FL is deployed in domains such as healthcare and finance where data sharing is legally constrained. **Limitation.** FL has not previously been applied to scientific-integrity screening, and scholarly-domain benchmarks with *naturally* non-IID silos are lacking; most FL benchmarks impose synthetic Dirichlet splits on image corpora.

Federated retraction prediction across publisher silos

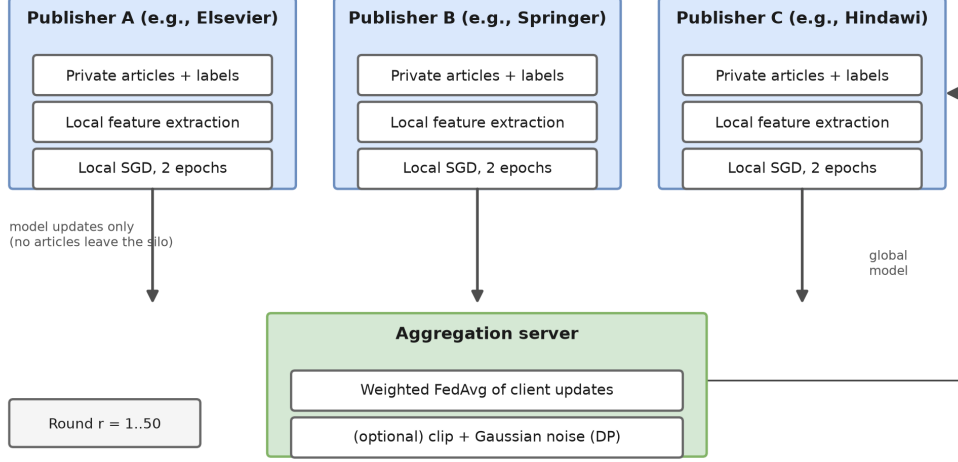


Figure 1: Federated retraction prediction. Publishers train locally on private articles; only model updates are aggregated (optionally clipped and noised for differential privacy), and the global model is redistributed.

3. Federated Retraction Prediction

3.1. Architecture

Figure 1 shows the system. Each publisher runs local feature extraction and local training on its private archive; a coordination server aggregates parameter updates into a global model that is redistributed each round. No article, label, or feature vector crosses a silo boundary, satisfying constraint (2).

3.2. Training procedure

We use the small multilayer perceptron f_θ (two hidden layers, 64 and 32 units, ReLU, dropout 0.2) for all federated experiments, so that federated

Algorithm 1 DP-enabled cross-silo FedAvg for retraction prediction

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1: input silos  $D_1, \dots, D_K$ ; rounds  $R$ ; local epochs  $E$ ; clip  $C$ ; noise  $\sigma$ 
2: server initializes  $\theta^{(0)}$ 
3: for  $r = 1, \dots, R$  do
4:   for each silo  $k = 1, \dots, K$  in parallel do
5:      $\theta_k \leftarrow \text{LOCALSGD}(\theta^{(r-1)}, D_k, E)$   $\triangleright$  minimizes (3); data stays in
       silo
6:      $\Delta_k \leftarrow \theta_k - \theta^{(r-1)}$ ;  $\Delta_k \leftarrow \Delta_k \cdot \min(1, C/\|\Delta_k\|_2)$ 
7:   end for
8:    $\bar{\Delta} \leftarrow \frac{1}{K} \sum_k \Delta_k + \mathcal{N}(0, \sigma^2 C^2 / K)$   $\triangleright \sigma=0$  recovers FedAvg
9:    $\theta^{(r)} \leftarrow \theta^{(r-1)} + \bar{\Delta}$ 
10: end for
11: return  $\theta^{(R)}$ 

```

and centralized results are architecture-identical. Local training minimizes class-weighted binary cross-entropy

$$\ell(\hat{y}, y) = -w_1 y \log \sigma(\hat{y}) - (1 - y) \log(1 - \sigma(\hat{y})), \quad w_1 = \frac{n_k^{(0)}}{n_k^{(1)}}, \quad (3)$$

with $n_k^{(c)}$ the count of class c in silo k . FedProx adds $\frac{\mu}{2} \|\theta - \theta^{(r)}\|^2$ to (3). Algorithm 1 summarizes one training run, including the optional differentially private aggregation: each client’s update is clipped to L_2 norm C and Gaussian noise $\mathcal{N}(0, \sigma^2 C^2 / K)$ is added to the average [11]; privacy is accounted over rounds with Rényi-DP composition.

3.3. Evaluation metrics

With retracted articles a minority class, accuracy is uninformative. We report the area under the precision–recall curve (AUPRC) as the primary

metric,

$$\text{AUPRC} = \int_0^1 \text{Prec}(\text{Rec}^{-1}(t)) dt, \quad (4)$$

whose random baseline equals the positive rate (≈ 0.27 on our corpus), together with ROC-AUC and recall at 5% false-positive rate (R@5%FPR), the fraction of future retractions caught when wrongly flagging only 5% of legitimate articles, the regime of an editorial pre-screening tool. All results are means $\pm$ standard deviation over three seeds.

4. A 138K-Article Federated Benchmark

Corpus construction. Positives are all Retraction Watch [1] research articles with a resolvable DOI whose record nature is *Retraction* (37,648 after filtering). Each is enriched via the OpenAlex API [12] with authorship, venue, citation, and abstract metadata. Controls are sampled at 3:1 from OpenAlex among never-retracted articles of the *same publisher and publication year* (cells with fewer than five positives are dropped), yielding 100,889 controls and 138,537 articles in total. Matching prevents the classifier from exploiting publisher- or era-priors instead of article-level signals.

Features. Each article is represented by $d=15$ features in two families, chosen to mirror the signal types validated in prior work [6, 4]: *metadata* (author count, country count, international collaboration, reference count, total citations, citations within two years of publication, publication year, open-access flag) and *abstract linguistics* (Flesch reading ease, Flesch–Kincaid grade, Gunning–Fog, SMOG, abstract length, hedging ratio, certainty ratio [13, 14]). In a real deployment each publisher would additionally compute

Table 1: Publisher silos (10-silo partition). Retraction rate is $\approx 25\%$ by matched design; Hindawi exceeds it where control sampling saturated.

Silo	Articles	Retracted	Retr. rate	Median year
Other (aggregated tail)	31,024	8,007	0.258	2018
Hindawi	27,696	9,277	0.335	2022
Elsevier	21,221	5,722	0.270	2017
Springer	19,730	4,930	0.250	2020
Wiley	13,372	3,340	0.250	2020
IOS Press	8,320	2,080	0.250	2022
PLOS	4,960	1,240	0.250	2016
SAGE	4,772	1,193	0.250	2019
Taylor & Francis	4,627	1,156	0.250	2018
Spandidos	2,815	703	0.250	2017

full-text features locally; these are exactly the signals shown to be strongest in prior work and inaccessible centrally; our public-metadata features are thus a conservative lower bound on federated performance.

Silos. Articles are partitioned into $K=10$ silos by publisher (Table 1) and, alternatively, by OpenAlex discipline. Silos differ strongly in size (2,815–31,024 test+train articles), median era (2016–2022), and composition. This is *natural* statistical heterogeneity, in contrast to the synthetic Dirichlet splits common in FL benchmarks.

5. Experiments

We evaluate on a stratified 20% held-out test set shared by all training regimes. *Local-only* trains one model per silo and averages test performance (the current deployable reality); *centralized* pools all data (the legally unattainable ideal; logistic regression, Random Forest [6]-style, XGBoost, and the MLP); *federated* trains the MLP with Algorithm 1 ($R=50$, $E=2$, SGD with momentum 0.9, learning rate 0.05, $\mu=0.01$).

5.1. Main result: federation closes the silo gap

Table 2 and Fig. 2 give the central result. Local-only training barely beats the random baseline (AUPRC 0.315 vs. 0.27): individual publishers, however large, cannot learn the global phenomenology of retraction. FedAvg reaches 0.558: **92% of the architecture-identical centralized MLP** (0.609) and a **77% improvement over the status quo**, with FedProx statistically indistinguishable. The pattern replicates on discipline silos (FedAvg 0.585), showing robustness to how the world is partitioned. Fig. 3 shows both strategies converge within ~ 30 rounds.

5.2. Why scale matters

Fig. 4 varies training-set size for the centralized XGBoost: AUPRC rises monotonically from 0.403 (500 articles, the scale of prior hand-curated corpora) through 0.573 (8K) to 0.633 (110K) and has not saturated. Sub-thousand corpora forfeit more than a third of attainable performance, an independent motivation for training at the scale only a federation can reach.

Table 2: Main benchmark (mean $\pm$ std, 3 seeds). Random AUPRC $\approx$ 0.27.

Partition	Model	AUPRC	ROC-AUC	R@5%FPR
Publisher	Local-only (avg)	0.315 $\pm$.003	0.558	0.075
	FedAvg	0.558 $\pm$.011	0.765	0.273
	FedProx	0.556 $\pm$.012	0.763	0.272
	FedBal (focal)	0.542 $\pm$.011	0.748	0.256
	Central. LogReg	0.354 $\pm$.005	0.601	0.108
	Central. RF	0.598 $\pm$.002	0.784	0.318
	Central. MLP	0.609 $\pm$.009	0.790	0.322
	Central. XGBoost	0.633 $\pm$.005	0.804	0.351
Discipline	Local-only (avg)	0.326 $\pm$.001	0.571	0.082
	FedAvg	0.585 $\pm$.012	0.777	0.293

5.3. Feature families: accuracy vs. portability

On the full corpus (Fig. 5), metadata features dominate in-domain accuracy (AUPRC 0.593 alone vs. 0.381 for abstract linguistics alone; 0.633 combined), and XGBoost feature importance concentrates on citation dynamics (total citations 0.136, early citations 0.128), echoing the citation-centric findings of [4]. The cross-corpus experiments of Sect. 5.7, however, show the roles reverse under domain shift: linguistic signal transfers, metadata does not.

5.4. Robustness: consortium size and temporal deployment

Consortium size. Varying the partition from $K=5$ to $K=20$ publisher silos changes FedAvg AUPRC only within noise (0.574/0.558/0.560/0.558 for $K=5/10/15/20$): the federation does not degrade as more, smaller publishers join. *Temporal split.* Training on articles published up to 2018 (52,506) and

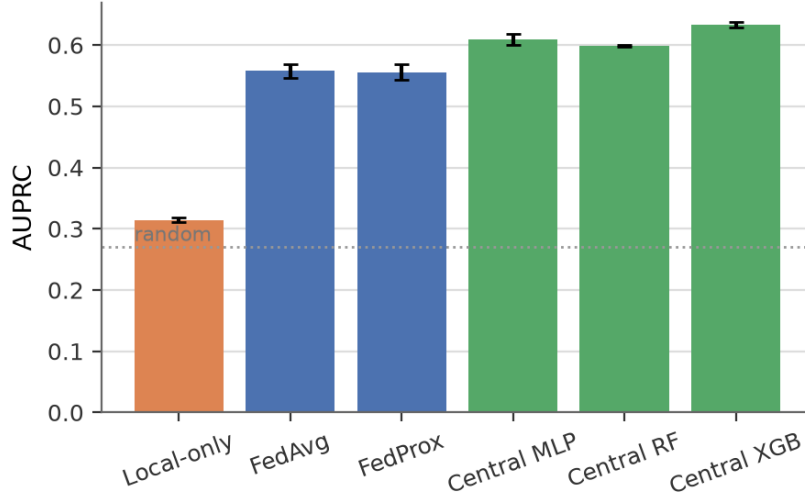


Figure 2: AUPRC by training regime, publisher partition (3 seeds). Federation (blue) recovers most of the unattainable centralized ideal (green) and far exceeds the silo-local status quo (orange).

testing on 2019 onward (86,031) is strictly harder for every regime (centralized XGBoost 0.436, MLP 0.424), reflecting distribution drift and retraction lag: many recent positives are not yet labeled. Critically, the federated model keeps pace (0.407, i.e., 93% of the centralized MLP ratio), so the cost of federation does not grow under realistic prospective deployment.

5.5. Per-silo effects and fairness

Evaluated on *their own* test slices, large publishers’ local models fit their in-distribution data well (e.g., Elsevier 0.708 local vs. 0.578 federated), while data-poor or atypical silos gain from federation (PLOS 0.419 vs. 0.386; Spanidos 0.449 vs. 0.425); see Fig. 7. Local specialization, however, is exactly what collapses on out-of-silo data (Sect. 5.1: local-only 0.315 globally). The federation thus buys universal coverage, benefits the weakest members most,

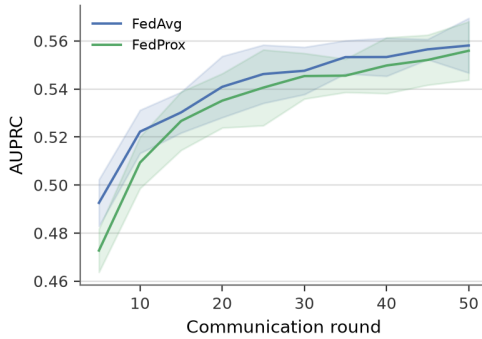


Figure 3: Federated convergence over communication rounds (publisher silos).

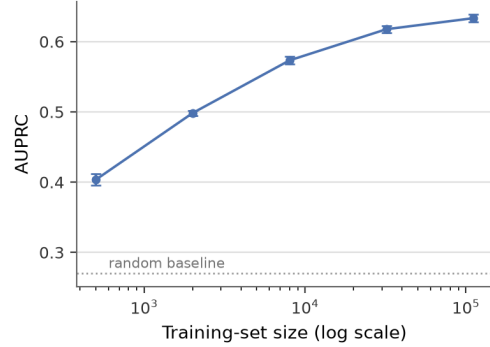


Figure 4: Centralized AUPRC vs. training-set size: signal grows monotonically with scale.

and motivates *personalized* FL (a global backbone with silo-specific heads) as future work.

5.6. Privacy-utility trade-off

Fig. 6 sweeps the noise multiplier $\sigma \in \{0.3, 0.5, 1.0, 2.0\}$ at clip $C=1$. Moderate noise ($\sigma=0.3$) retains 95% of the non-private AUPRC (0.514 vs. 0.540); $\sigma \geq 1$ collapses utility. Because only $K=10$ clients participate, the resulting formal ϵ values are large; with realistic consortium sizes (hundreds of publishers and journals) the same mechanism yields far stronger guarantees. We report the curve as an honest characterization of the mechanism at benchmark scale.

5.7. Cross-corpus transfer against the hand-curated benchmark

We obtained the publicly released corpus of [6] (232 retracted / 232 matched non-retracted biomedical articles with full-text sections). Table 3

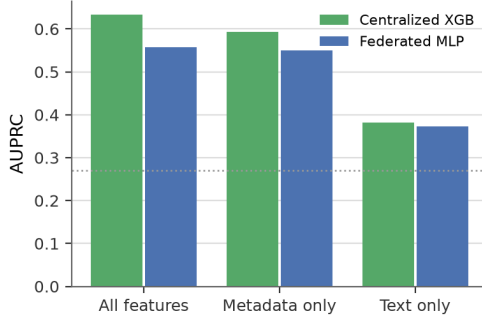


Figure 5: Feature-family ablation, centralized and federated.

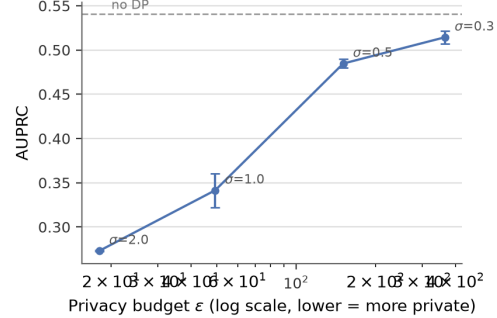


Figure 6: Privacy-utility trade-off of DP aggregation (Algorithm 1).

reports three experiments that jointly characterize the relationship between the small-corpus and large-corpus regimes.

Three findings follow. *First* (T1), a model trained on hundreds of curated articles collapses to chance on the open corpus: small-corpus results do not generalize, independently of feature quality. *Second* (T4), restricted to abstract-level features, the very architecture that achieves 0.88 accuracy with full-text sections [6] drops to chance on its own corpus; to our knowledge this is the first quantification of how much retraction signal resides exclusively in paywalled full text. This is the strongest empirical argument for federation: the decisive features exist only where the data cannot leave. *Third* (T2 vs. T3), our large-corpus model transfers to the curated biomedical benchmark only through its *linguistic* features (0.691); adding metadata features destroys transfer (0.508), since citation and era priors are domain-specific. Writing style is the portable component of retraction signal.

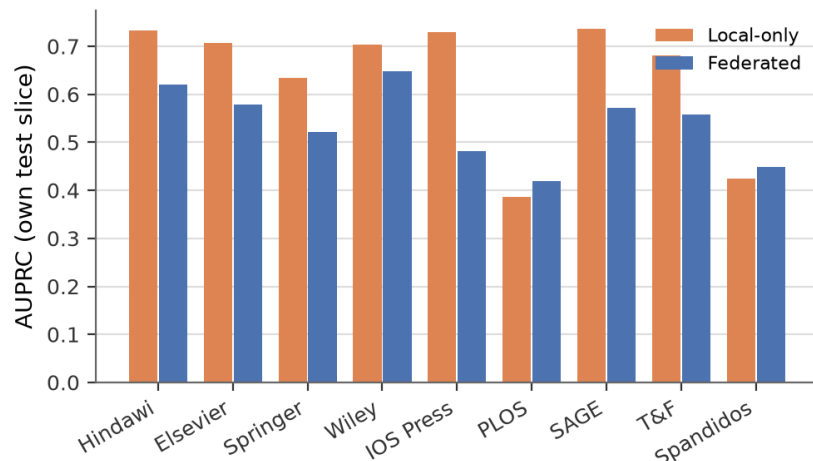


Figure 7: Local vs. federated AUPRC on each publisher’s own test slice. Federation helps data-poor silos; large silos specialize locally but fail globally (cf. Table 2).

6. Discussion and Limitations

Simulation vs. deployment. Our silos are simulated from public metadata; a production federation would add per-publisher full-text features, the signals T4 shows to be decisive, making our results a conservative lower bound. *Label noise.* Recent articles may be retracted after data collection; the temporal experiment bounds this effect. *Coverage bias.* Retraction Watch over-represents fields and publishers with active integrity teams; publisher-year matching removes the corresponding shortcut from the classifier but not from the label distribution. *Privacy formalism.* With $K=10$ clients, DP guarantees are loose; they tighten mechanically with consortium size. *Negative result.* A focal-loss, prevalence-weighted aggregation variant (FedBal, Table 2) did not outperform FedAvg because matched controls equalize prevalence across silos by construction; under unmatched deployment prevalence

Table 3: Cross-corpus experiments. Random AUPRC: 0.50 on the balanced corpus of [6], 0.27 on ours.

#	Experiment	AUPRC	Baseline
T1	RF trained on 464 curated articles $\rightarrow$ our 138K corpus	0.283	0.27
T2	Our federated model (all features) $\rightarrow$ curated corpus	0.508	0.50
T3	Our model, <i>linguistic features only</i> $\rightarrow$ curated corpus	0.691	0.50
T4	RF on curated corpus itself, abstract- only features (5-fold CV)	0.525	0.50

skew it remains an open question.

7. Conclusion

We reformulated retraction prediction as cross-silo federated learning and showed, on a corpus $180\times$ larger than prior work, that publishers can jointly train a screening model that recovers 92% of the unattainable centralized ideal without sharing a single record, while models trained on any silo alone, or on small curated corpora, fail outside their origin. By quantifying the full-text signal locked behind paywalls ($0.88 \rightarrow$ chance without it) and the portability of linguistic features, our results chart the path to integrity screening at the scale of the scientific record: a publisher federation, linguistically grounded features, and privacy mechanisms whose guarantees strengthen exactly as the federation grows. For the digital-library and scholarly-communication community, this reframes retraction screening from

a data-pooling problem, which is legally foreclosed, into a model-sharing one, which is not.

CRedit authorship contribution statement

Muhammad Ahmad: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft. **Muhammad Tanvir Afzal:** Supervision, Conceptualization, Validation, Writing – review & editing.

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Data availability

The corpus is built from open sources: the Retraction Watch database (distributed via Crossref), the OpenAlex API, and Europe PMC. The corpus-construction pipeline, feature extraction, the model-ready feature matrix, and all experiment configurations are released to enable full reproduction at <https://github.com/Ahmad-prog/federated-retraction-prediction>.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI in the writing process

The author used AI-assisted tools for language editing and LaTeX formatting. All scientific content, experiments, analyses, and conclusions are the author's own, and the author reviewed and takes full responsibility for the content of the manuscript.

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